

A multilevel $\mathcal{O}(N \log N)$ algorithm for evaluating the attenuated Radon transform

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September 26, 2026

Abstract

The forward operator of emission tomography is the attenuated Radon transform, in which the attenuation enters through an exponential whose argument varies along the line of integration. Evaluating it on an $n \times n$ grid for $\mathcal{O}(n)$ directions costs $\mathcal{O}(n^3) = \mathcal{O}(N^{3/2})$ operations by direct quadrature, where $N = n^2$, and this evaluation is the inner loop of every iterative reconstruction method. We give an $\mathcal{O}(N \log N)$ algorithm in the multilevel integral-transform framework of Brandt and Lubrecht. The algorithm rests on an exact concatenation identity for a triple of segment integrals, which lets the length of the integration segment double from level to level with no approximation, and on the observation that a segment of length ℓ resolves direction only to $\delta\theta \sim h/\ell$, so that short segments require few directions. Angular resolution is therefore doubled at the finest levels in step with the segment length and the coarsest levels perform exact merges only. We identify the quantities that must be interpolated — the attenuation exponent rather than its exponential, and a segment integral referenced to its own exit point rather than to its midpoint — and show that the alternatives are unstable. Numerical experiments on a battery of attenuation and activity phantoms confirm the predicted cost and show that the error introduced by the multilevel schedule is an order of magnitude below the discretisation error of the underlying grid by a factor of six to twelve, at a cost already $2.9\times$ lower in operations and $4.5\times$ lower in wall time at $n = 129$, with the ratio falling as $\mathcal{O}(\log N/\sqrt{N})$.

1 Introduction

1.1 The forward model

Emission tomography reconstructs the spatial distribution of a radiotracer from counts of the photons it emits. Two fields are involved: the activity $f(\mathbf{x}) \geq 0$, which is the unknown, and the linear attenuation coefficient $\mu(\mathbf{x}) \geq 0$ of the surrounding medium at the photon energy, which is usually measured separately by a transmission or CT scan.

Parameterise a line in $\mathbb{R}^2$ by a direction and a signed offset,

$$\boldsymbol{\omega}(\theta) = (\cos \theta, \sin \theta), \quad \boldsymbol{\omega}^\perp(\theta) = (-\sin \theta, \cos \theta), \quad \mathbf{x}(s, t) = s \boldsymbol{\omega}^\perp + t \boldsymbol{\omega}, \quad (1)$$

and traverse $L(\theta, s) = \{\mathbf{x}(s, t) : t \in \mathbb{R}\}$ in the direction of increasing t . A photon emitted at $\mathbf{x}(s, t)$ and travelling along $\boldsymbol{\omega}$ escapes with probability $\exp(-\int_t^\infty \mu)$, so the expected count along $L(\theta, s)$ is

$$(\mathcal{R}_\mu f)(\theta, s) = \int_{\mathbb{R}} f(\mathbf{x}(s, t)) \exp\left(-\int_t^\infty \mu(\mathbf{x}(s, \tau)) d\tau\right) dt. \quad (2)$$

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This is the attenuated Radon transform, also called the attenuated X-ray transform. Its distinguishing feature is that the exponent depends on the emission point t , so the attenuation does not factor out of the integral. Equation (2) is the forward model of single-photon emission computed tomography (SPECT).

Positron emission tomography (PET) detects two collinear annihilation photons in coincidence. Both must escape, in opposite directions, so the survival probability is $\exp(-\int_t^\infty \mu) \exp(-\int_{-\infty}^t \mu) = \exp(-\int_L \mu)$, independent of t , and

$$(\mathcal{P}_\mu f)(\theta, s) = \exp\left(-\int_L \mu\right) \int_L f. \quad (3)$$

Classical PET attenuation is thus a multiplicative factor per line. The algorithm below computes the three line integrals

$$S = \int_L f, \quad E = \int_L \mu, \quad I = (\mathcal{R}_\mu f)(\theta, s), \quad (4)$$

and therefore delivers both models: (2) is I , and (3) is $e^{-E}S$. The exponent is genuinely position-dependent only in (2), and that is the case treated here; (3) is recovered as a by-product at no extra cost. Time-of-flight PET, in which the coincidence is localised along the line and the two one-sided attenuations no longer combine into a constant, has the structure of (2) and is covered by the same machinery.

1.2 The inverse problem and where evaluation enters

With μ known, (2) is a linear Fredholm integral equation of the first kind for f ,

$$\mathcal{R}_\mu f = g, \quad (5)$$

where g is the measured sinogram. Novikov [12] gave an explicit inversion formula for $\mathcal{R}_\mu$, and for the unattenuated case $\mu \equiv 0$ the classical filtered-backprojection inversion of the Radon transform applies [11]. In practice (5) is solved iteratively rather than by an inversion formula, because the data are Poisson counts, because the system is large and ill-conditioned, and because regularisation and physical corrections are easier to impose inside an iteration.

The standard methods are all of this kind. The expectation-maximisation algorithm for the Poisson likelihood [13] updates

$$f^{(k+1)} = \frac{f^{(k)}}{\mathcal{R}_\mu^* \mathbf{1}} \mathcal{R}_\mu^* \left(\frac{g}{\mathcal{R}_\mu f^{(k)}} \right), \quad (6)$$

a multiplicative — hence nonlinear — relaxation; its ordered-subset variant [10] applies (6) over subsets of directions in sequence, which makes it a block row-action scheme in the family of algebraic reconstruction techniques [8], i.e. a Kaczmarz relaxation with a nonlinear update. Every sweep of every one of these methods applies $\mathcal{R}_\mu$ once and its adjoint once. The cost of the reconstruction is therefore a small multiple of the cost of evaluating $\mathcal{R}_\mu$, and an $\mathcal{O}(N^{3/2})$ evaluation caps the resolution that is reachable.

Two further considerations motivate a multilevel evaluation specifically rather than any fast evaluation. First, if μ is not measured separately but estimated jointly with f from the emission data, the problem becomes nonlinear in μ , since μ sits in the exponent; the natural solver is then a full approximation scheme (FAS) [3] in which the nonlinearity is handled on a hierarchy

of grids. Second, a row-action relaxation of (5) is a smoother in the multigrid sense: it reduces the components of the residual that the fine grid resolves, and leaves the smooth components to be handled on coarser grids, so a full multigrid (FMG) cycle over the image hierarchy is the natural outer solver. The level hierarchy constructed here — segment lengths paired with angular resolutions — is the same hierarchy such a solver would traverse, which is what makes the evaluation and the solver composable. We do not develop the solver in this paper. The present contribution is the forward evaluation; the adjoint, the nonlinear inversion and the FAS/FMG cycle are the subject of subsequent work (Section 6).

1.3 Related work and contribution

Fast summation by separating a smooth kernel on a hierarchy of grids is the subject of the multilevel matrix-multiplication framework of Brandt and Lubrecht [6] and its extension to oscillatory kernels [4]; the fast multipole method [9] solves the same problem by a different decomposition. For line integrals specifically, Brandt and Dym [5] gave an $\mathcal{O}(N \log N)$ algorithm based on recursive doubling of segment length, and Brandt et al. [7] built a multilevel inversion of the unattenuated Radon transform on it. Fast backprojection by hierarchical decomposition [2] and transforms built on the pseudopolar grid [1] achieve comparable asymptotics for the unattenuated transform.

The contribution here is the extension to the attenuated transform (2), where the smoothness of the exponent and of the integrand must be exploited together. Specifically:

1. an exact concatenation identity (Proposition 1) for a triple of segment integrals, which makes the doubling of segment length exact and, in addition, damping: an error committed upstream is multiplied by $e^{-E} \leq 1$ at every merge;
2. identification of the quantities that admit interpolation — the exponent E rather than e^{-E} , and an exit-referenced rather than midpoint-referenced attenuated integral — with the failure mode of each alternative;
3. a sampling analysis of the three axes (direction, transverse offset, position along the ray) which fixes the level schedule and shows that the angular refinement must be placed at the finest levels, not the coarsest;
4. a measured account of the resulting accuracy and cost, including the finding that the error budget is dominated not by the angular interpolation the method is built around, but by the resampling of the tables in the transverse and along-ray variables when the ray frame rotates.

An open-source reference implementation and the scripts that produce every number and figure below are available at <https://github.com/orenlivne/tomogrid>.

2 Segment integrals and the concatenation identity

The algorithm works with integrals over finite segments of a ray rather than over whole lines. Fix (θ, s) and write $f(t)$, $\mu(t)$ for the restrictions to that ray. For $a < b$ define

$$E[a, b] = \int_a^b \mu(\tau) d\tau, \quad S[a, b] = \int_a^b f(t) dt, \quad I[a, b] = \int_a^b f(t) e^{-\int_t^b \mu(\tau) d\tau} dt. \quad (7)$$

The attenuation in I is referenced to the segment's own exit point b , so every weight lies in $(0, 1]$. This choice is what makes the recursion stable (Remark 1).

Proposition 1 (Concatenation). *For any $c \in (a, b)$,*

$$E[a, b] = E[a, c] + E[c, b], \quad S[a, b] = S[a, c] + S[c, b], \quad I[a, b] = e^{-E[c, b]} I[a, c] + I[c, b]. \quad (8)$$

Proof. The first two are additivity of the integral. For the third, split at c . For $t \in [a, c]$ write $\int_t^b \mu = \int_t^c \mu + \int_c^b \mu$; the second term does not depend on t and factors out as $e^{-E[c, b]}$, leaving $I[a, c]$. For $t \in [c, b]$ the integrand is unchanged and contributes $I[c, b]$. $\square$

Equivalently, the lower-triangular matrices

$$M[a, b] = \begin{pmatrix} e^{-E[a, b]} & 0 \\ I[a, b] & 1 \end{pmatrix} \quad \text{satisfy} \quad M[a, b] = M[a, c] M[c, b], \quad (9)$$

so the triple (S, E, I) carries a semigroup structure under concatenation. Identity (8) involves no quadrature and no interpolation: given the triples of two adjacent segments, the triple of their union is obtained exactly. This is what permits the segment length to double from level to level at no cost in accuracy.

Remark 1 (Why I is referenced to the exit point). *Referencing the attenuation to the segment midpoint m , i.e. using $\tilde{I}[a, b] = \int_a^b f(t) e^{-\int_t^m \mu dt}$, gives weights as large as $e^{+E[a, m]}$, which grow without bound with segment length and attenuation strength. With the exit-referenced definition (7) all weights are in $(0, 1]$, and (8) shows that an error ε in $I[a, c]$ propagates into $I[a, b]$ as $e^{-E[c, b]}\varepsilon$. Strong attenuation therefore improves the conditioning of the recursion. This is confirmed in Section 5.2: the phantom with the largest attenuation gives the smallest error.*

Remark 2 (Why E and not e^{-E} is interpolated). *E is an additive line integral of μ , smoothed along the ray by the averaging in (7), with derivatives bounded in terms of those of μ . Its exponential spans many orders of magnitude — in the experiments below E reaches 12, so $e^{-E} \approx 6 \times 10^{-6}$ — and has correspondingly large relative derivatives. The algorithm interpolates E and applies the exponential afterwards. This is the sense in which smoothness is used twice: once in the exponent, through E , and once in the integrand, through I .*

3 Smoothness and sampling

Let $Q(\theta)$ denote a segment integral of length ℓ centred at a fixed point $\mathbf{x}$, as a function of the segment's direction. For a single Fourier mode $f = e^{i\mathbf{k} \cdot \mathbf{y}}$ with $|\mathbf{k}| = k$,

$$Q(\theta) = \int_{-\ell/2}^{\ell/2} e^{i\mathbf{k} \cdot (\mathbf{x} + \tau \boldsymbol{\omega}(\theta))} d\tau = e^{i\mathbf{k} \cdot \mathbf{x}} \ell \operatorname{sinc}\left(\frac{\ell k}{2} \cos(\theta - \phi)\right), \quad \phi = \arg \mathbf{k}. \quad (10)$$

The θ -dependence enters only through $u(\theta) = \frac{\ell k}{2} \cos(\theta - \phi)$, with $|du/d\theta| \leq \ell k/2$, and sinc varies on unit scale in u . Hence Q varies in θ on the scale $\delta\theta \sim 2/(\ell k)$. With the finest scale representable on a grid of spacing h being $k_{\max} \sim \pi/h$, the angular sampling requirement is

$$\boxed{\delta\theta_\ell \ell \lesssim c h, \quad c = \mathcal{O}(1).} \quad (11)$$

Geometrically (11) says that the endpoints of the segment must not move by more than about one grid cell between neighbouring directions. Short segments need few directions.

The other two axes follow from the same expression. Transverse to the ray no averaging takes place, so the sample spacing δs must resolve f itself. Along the ray, Q is a box average of length

ℓ , whose transfer function is $\text{sinc}(k\ell/2)$; this is not a strong low-pass filter, since at the Nyquist frequency $k = \pi/\ell$ of a grid with $\delta t = \ell$ it still equals $2/\pi \approx 0.64$. Sampling the tables at $\delta t = \ell$ therefore aliases. Taking

$$\delta t = \ell/\nu, \quad \nu = 2, \quad (12)$$

places the grid Nyquist frequency exactly at the first zero of $\text{sinc}(k\ell/2)$. Section 5.5 confirms that $\nu = 2$ is both necessary and sufficient: $\nu = 4$ and $\nu = 8$ give no further improvement. Segments at a given level consequently overlap by half.

4 The algorithm

4.1 Hierarchy

Let the support of f and μ lie in the disc of radius R and let $H > R$ be a half-chord that covers any line of interest. Levels $l = 0, \dots, L$ carry segments of length

$$\ell_l = 2^l \ell_0, \quad \ell_0 \approx h, \quad \ell_L = 2H. \quad (13)$$

Level l stores, for each direction θ_k on a grid Θ_l , a table of the triple (7) over (s, t) , where t is the segment centre. The grids are

$$\delta s = h/\sigma, \quad \delta t_l = \ell_l/\nu, \quad |\Theta_l| = \min(n_\theta, n_\theta^{\min} 2^l), \quad (14)$$

with $\Theta_l = \{2\pi i/|\Theta_l|\}$ so that $\Theta_{l-1} \subset \Theta_l$, and t -grids $t_j = -H + j \delta t_l$ so that the level- l grid is every second node of the level- $(l-1)$ grid.

The last entry of (14) is the scheduling rule, and its direction is the point that is easy to get backwards. By (11) long segments need many directions and short ones few, so the angular grid is doubled at the *finest* levels, in step with the doubling of ℓ_l . While the doubling is active,

$$\delta \theta_{l-1} \ell_{l-1} = \frac{2\pi}{n_\theta^{\min} 2^{l-1}} \cdot \ell_0 2^{l-1} = \frac{2\pi \ell_0}{n_\theta^{\min}}, \quad (15)$$

independent of l : every refinement faces the same criterion, and with $\ell_0 \approx h$ the constant in (11) is $c = 2\pi/n_\theta^{\min}$. A single parameter, $n_\theta^{\min}$, controls the angular interpolation error. Once (14) saturates at n_θ , the remaining levels perform exact merges only and contribute no error at all. Setting $n_\theta^{\min} = n_\theta$ disables every interpolation and recovers direct evaluation, which is how the reference solutions in Section 5 are computed.

4.2 Level transition

Each level transition consists of an angular prolongation followed by a merge.

(A) Angular prolongation. $\Theta_{l-1} \subset \Theta_l$, so half the directions of Θ_l are copies of level- $(l-1)$ directions and are taken over exactly. Each new direction θ is the midpoint of two old ones, and its table is obtained by interpolation in θ over a stencil of q old directions. The interpolation must be performed at a fixed *spatial* point: the value of the old table at direction θ' is needed at $s' = \mathbf{x} \cdot \boldsymbol{\omega}^\perp(\theta')$, $t' = \mathbf{x} \cdot \boldsymbol{\omega}(\theta')$ where $\mathbf{x} = s \boldsymbol{\omega}^\perp(\theta) + t \boldsymbol{\omega}(\theta)$, which requires a two-dimensional interpolation in the old frame's (s, t) grid. Interpolating in θ at fixed *ray* coordinates instead would rotate the segment centre on a circle of radius $|\mathbf{x}| = \mathcal{O}(1)$ and, by (10), force $\delta \theta \lesssim h$ at every level, destroying the complexity.

Algorithm 1 Multilevel evaluation of (4)

- 1: build level-0 tables by (16) on $\Theta_0 \times \{s\} \times \{t\}$
 - 2: **for** $l = 1, \dots, L$ **do**
 - 3: **(A)** copy tables at Θ_{l-1} ; interpolate the remaining directions of Θ_l in (θ, s, t)
 - 4: **(B)** merge node pairs $2j \mp \nu/2$ by (8)
 - 5: **end for**
 - 6: **return** the level- L table at $t = 0$
-

(B) Merge. At each direction, the level- l segment centred at level- $(l-1)$ node index $2j$ is the union of the level- $(l-1)$ segments centred at $2j \mp \nu/2$; the offset $\ell_{l-1}/2$ is an integer number of grid steps precisely because ν is even. The triples are combined by (8). No interpolation in t occurs, and the step is exact.

4.3 Finest level

Level-0 segments have length $\approx h$ and are evaluated directly. Place m Gauss–Legendre nodes $\{t_i\}$ with weights $\{w_i\}$ on the segment, sample f and μ there by tensor-product interpolation from the image grid, and take

$$E = \frac{\ell}{2} \sum_j w_j \mu_j, \quad S = \frac{\ell}{2} \sum_j w_j f_j, \quad A_i = \frac{\ell}{2} \sum_j C_{ij} \mu_j, \quad I = \frac{\ell}{2} \sum_i w_i f_i e^{-A_i}, \quad (16)$$

where $C_{ij} = \int_{t_i}^b L_j$ and L_j is the Lagrange basis on the nodes. A_i approximates $\int_{t_i}^b \mu$ to order m from the same m samples, so the partial attenuations needed by I cost one $m \times m$ matrix application per segment. Note that (16) is accurate only while $\mu\ell$ is moderate — an 8-node rule applied to a whole chord with $E = 30$ has relative error 10^{-3} — which is precisely why the direct quadrature is confined to the shortest segments and everything above is the exact recursion (8).

4.4 Zero extension

Outside the tables the triples are taken to be zero. This is exact rather than an approximation: if $|s'| > R$ the whole line misses the support, and if $|t'| > H \geq R + \ell/2$ the segment does. The interpolation stencils may therefore straddle the table boundary without loss, provided f and μ vanish smoothly; a field with a jump violates the smoothness on which every interpolation in the method depends, and the order degrades accordingly (Section 5.5).

4.5 Complexity

Work at level l is proportional to the table size $|\Theta_l| n_s n_t^{(l)}$ with $n_t^{(l)} = \nu 2^{L-l} + 1$. Writing $M = 2^L \approx 2H/h$ and $l^* = \log_2(n_\theta/n_\theta^{\min})$,

$$|\Theta_l| n_t^{(l)} \approx \begin{cases} n_\theta^{\min} \nu M, & l \leq l^* \quad (\text{refining}), \\ n_\theta \nu M / 2^l, & l > l^* \quad (\text{coasting}). \end{cases} \quad (17)$$

The refining levels each cost the same and the coasting levels decay geometrically, so

$$W_{\text{ML}} = \mathcal{O}(n_\theta^{\min} \nu M n_s (l^* + 2)) = \mathcal{O}(N \log N), \quad N = n^2 \approx M n_s, \quad (18)$$

against $W_{\text{direct}} = \mathcal{O}(n_\theta n_s M) = \mathcal{O}(N^{3/2})$. The constant per refining level is $q p_s p_t$ table lookups, where p_s, p_t are the stencil widths.

5 Numerical results

All experiments use the reference implementation in `tomogrid`: cubic (4-point) stencils throughout, $m = 8$ Gauss–Legendre nodes per level-0 segment, $\nu = 2$, $\sigma = 2$, $n_\theta^{\min} = 16$ unless stated, $H = 1$, images on $[-1, 1]^2$. Directions are uniform on $[0, 2\pi)$; the full circle is required because (2) is not invariant under $\theta \mapsto \theta + \pi$, unlike S and E .

Three errors are reported, all measured in the supremum norm over the sinogram and normalised by $\max |S|$, which is the physically meaningful scale since I itself underflows on heavily attenuated rays:

- e_{alg} : multilevel vs direct evaluation on the *same* discretisation;
- e_{disc} : direct evaluation vs the continuum;
- e_{tot} : multilevel vs the continuum.

Separating them matters: e_{alg} is what the multilevel schedule costs, and it is the only quantity the algorithm controls.

5.1 Verification

The continuum reference is a dense trapezoidal evaluation of the analytic phantom. On the C^∞ compactly supported fields used here every Euler–Maclaurin boundary term vanishes and the rule is spectrally accurate: against the closed-form Radon transform of a Gaussian with uniform attenuation it agrees to 2×10^{-14} , and doubling its sample count changes the result by 8.3×10^{-8} .

The exactness claims are verified independently. Proposition 1 reproduces whole-segment triples from split ones to 10^{-13} ; the nested t -grid bookkeeping of step (B) reproduces directly computed coarse-level tables to 10^{-12} for $\nu \in \{2, 4, 8\}$; and the physical limits hold to roundoff — with $\mu \equiv 0$ the computed I equals S to 10^{-12} , and when the attenuator lies entirely downstream of the source the SPECT model collapses onto the PET model $e^{-E}S$ to 10^{-10} , while for the reversed direction $I = S$ to the same tolerance.

5.2 Attenuation battery

Table 1 reports e_{alg} over nine attenuation fields crossed with four activity fields on a deliberately coarse 33^2 grid with $n_\theta^{\min} = 8$, where the method is under maximum stress. The fields include a uniform soft-edged disc, single and multiple blobs, a high-contrast case, an anisotropic blob with large $|\nabla\mu|$ in one direction, a linear gradient, a short-wavelength oscillatory field, and a case with chord attenuation $E \approx 12$.

The error varies by a factor of six across the battery, and short-wavelength μ is no worse than smooth μ . The strongly attenuating case gives the *smallest* error in I , as predicted by Remark 1: the factor $e^{-E[c,b]}$ in (8) damps upstream error. The worst case is the high-contrast field, consistent with the interpolation constant growing with $|\nabla\mu|$.

5.3 Refinement

Table 2 and Figure 1 report the refinement study: a fixed phantom (three blobs through a high-contrast attenuator, peak chord attenuation $E \approx 3.1$) on grids $n = 33, 65, 129, 257$ with $n_\theta = n - 1$, $n_s = n$ and $n_\theta^{\min} = 16$ held fixed.

Two observations. First, e_{alg} is a factor of six to twelve below e_{disc} at every resolution, so the multilevel schedule is not what limits the accuracy of the computed sinogram; the discretisation of

Table 1: Algorithmic error e_{alg} in I over the phantom battery, on a 33^2 grid with $n_\theta = 32$ and $n_\theta^{\min} = 8$, i.e. under maximum stress. Columns are activity fields, rows attenuation fields; $\max E$ is the largest chord attenuation. Worst entry 3.0×10^{-3} .

attenuation	$\max E$	centered_bump	offcenter_bump	three_blobs	smooth_phantom
zero	0.00	6.5×10^{-4}	2.1×10^{-3}	2.7×10^{-3}	2.1×10^{-3}
uniform_soft	1.02	5.1×10^{-4}	1.4×10^{-3}	2.0×10^{-3}	1.6×10^{-3}
single_bump	1.21	6.6×10^{-4}	1.3×10^{-3}	2.0×10^{-3}	1.8×10^{-3}
two_bumps	1.38	1.2×10^{-3}	2.3×10^{-3}	2.3×10^{-3}	1.8×10^{-3}
high_contrast	3.08	2.1×10^{-3}	3.0×10^{-3}	2.4×10^{-3}	1.8×10^{-3}
anisotropic	1.99	1.2×10^{-3}	2.7×10^{-3}	2.8×10^{-3}	2.0×10^{-3}
gradient	1.57	5.4×10^{-4}	1.1×10^{-3}	1.9×10^{-3}	1.5×10^{-3}
oscillatory	1.73	5.7×10^{-4}	1.3×10^{-3}	2.4×10^{-3}	1.7×10^{-3}
very_strong	12.02	2.0×10^{-4}	9.5×10^{-4}	8.3×10^{-4}	4.1×10^{-4}

Table 2: Refinement study. Activity **three_blobs** through attenuation **high_contrast** (peak chord attenuation $E \approx 3.1$), with $n_\theta = n - 1$, $n_s = n$, $n_\theta^{\min} = 16$, $\sigma = \nu = 2$. Errors are relative sup errors; p is the observed order between consecutive rows. Direct evaluation at $n = 257$ was not run; its operation count is from the schedule. Times are single-threaded NumPy on one core and are indicative only.

n	n_θ	e_{alg}		e_{disc}		e_{tot}		Gflop		seconds	
		err	p	err	p	err	p	ML	direct	ML	direct
33	32	2.05×10^{-3}	—	2.45×10^{-2}	—	2.42×10^{-2}	—	0.07	0.09	0.5	1.3
65	64	4.50×10^{-4}	2.19	5.97×10^{-3}	2.04	5.92×10^{-3}	2.03	0.38	0.68	3.8	9.0
129	128	1.38×10^{-4}	1.70	7.74×10^{-4}	2.95	7.65×10^{-4}	2.95	1.90	5.41	15.4	69.0
257	256	—	—	—	—	7.67×10^{-5}	3.32	9.20	43.12	77.8	—

the image is. Second, both errors converge under refinement, which is the property that makes the method usable: the multilevel error is not a fixed tolerance but goes to zero with the mesh.

The observed orders are pre-asymptotic on this phantom, which contains a blob of radius 0.22 that is under-resolved at $n = 33$. On a well-resolved smooth phantom (a single C^∞ bump of radius 0.7) the discretisation error converges at observed orders 3.03, 3.42, 3.53 over the same grid sequence, approaching the fourth order of the cubic image interpolant. The residual defect is attributable to the C^0 kinks of that interpolant, which the level-0 quadrature straddles: raising the interpolation stencils from fourth to sixth order does not improve the rate, whereas a C^2 image representation would be expected to.

5.4 Cost

The measured cost in Table 2 follows (18). Direct evaluation grows by a factor of 8 per grid doubling and the multilevel algorithm by a factor between 4 and 6, so the advantage grows as $\mathcal{O}(\sqrt{N}/\log N)$. At $n = 129$ the operation count is $2.9\times$ lower and the wall time $4.5\times$ lower; the gap between the two ratios is a memory effect, the direct schedule having a much larger level-0 working set. Table 3 extrapolates the operation count, which is exact arithmetic on the schedule rather than a measurement.

The crossover is early — the multilevel algorithm is already cheaper at $n = 65$ — because its overhead is a small constant times the level-0 work rather than an asymptotic-only saving.

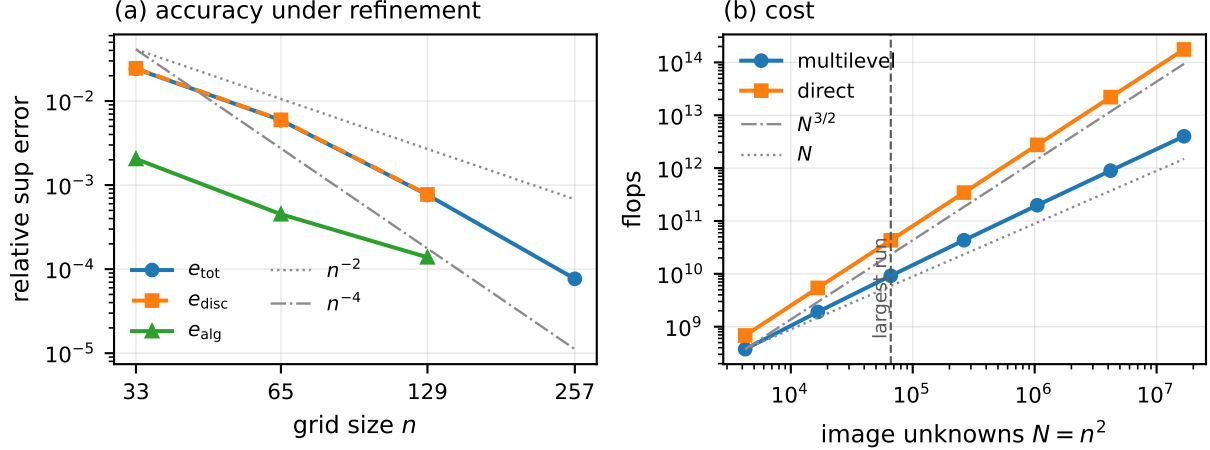


Figure 1: (a) Relative sup errors against grid size, from Table 2, with reference slopes n^{-2} and n^{-4} . The algorithmic error e_{alg} stays well below the discretisation error e_{disc} , which is indistinguishable from the total error e_{tot} at this scale, and all converge. (b) Operation count against the number of image unknowns $N = n^2$ from the schedule of Section 4.5, with reference slopes N and $N^{3/2}$; the dashed line marks the largest case actually executed.

Table 3: Operation count from the schedule of Section 4.5, with $n_\theta = n - 1$, $n_s = n$, $n_\theta^{\min} = 16$. Exact arithmetic on the level structure, not a measurement.

n	multilevel (Gflop)	direct (Gflop)	ratio
65	0.4	0.7	1.8
129	1.9	5.4	2.8
257	9.2	43.1	4.7
513	43.2	344.3	8.0
1025	198.5	2751.5	13.9
2049	896.8	22001.0	24.5
4097	3999.1	175964.8	44.0

5.5 Parameters

Table 4 isolates the accuracy parameters at $n = 129$.

$n_\theta^{\min}$ behaves as (15) predicts, and $n_\theta^{\min} = n_\theta$ gives zero error by construction. $\nu = 2$ is confirmed optimal: $\nu = 4$ and $\nu = 8$ cost two and four times as much and change the error by under 2%, as (12) predicts. The transverse oversampling σ is the dominant knob, and Figure 2 shows it is also the most efficient one: at equal cost it buys more accuracy than any of the others.

The stencil widths in the lower block of Table 4 locate the error budget, and the answer is not what the design would suggest. Narrowing the angular stencil from fourth to second order costs a factor of 1.9, whereas narrowing the transverse and along-ray stencils costs factors of 14 and 6.5; widening the angular stencil to sixth order gains 2.5% against 23% for the transverse one. The dominant term is therefore not the interpolation in θ — the operation the method is built around — but the resampling of the tables in (s, t) that accompanies it, a consequence of storing the tables in ray-aligned coordinates rather than an intrinsic property of the multilevel decomposition. A formulation that avoids the resampling, for instance by carrying frame-covariant quantities, would improve on what is reported here.

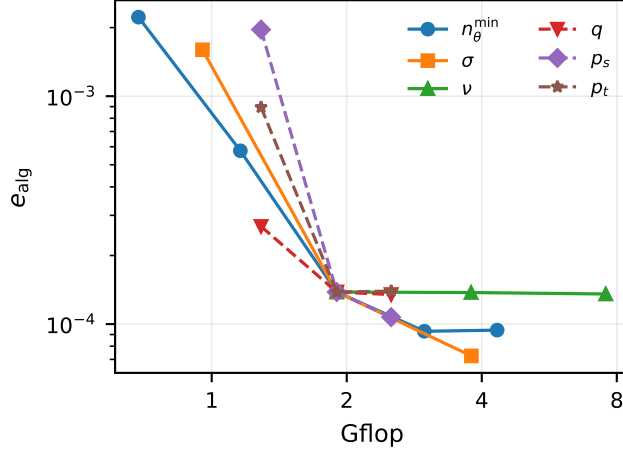


Figure 2: Algorithmic error against operation count at $n = 129$ as each parameter of Table 4 is varied in turn (solid: sampling parameters; dashed: stencil widths). The transverse oversampling σ is the most efficient direction to spend work in; ν beyond 2 buys nothing; and the angular stencil q moves the error far less than the transverse one p_s .

Finally, the smoothness assumption is not decorative. Replacing the smooth activity by a discontinuous disc raises e_{alg} by a factor of 3.5 at the settings of Table 1 and, more tellingly, the error then stops converging under refinement.

6 Discussion

Limitations. Three are worth stating precisely. (i) On rays with very large total attenuation the true signal underflows, and what the method controls is a small error *absolute* to $\max |S|$, not a small relative error; carrying the mean survival weight $W = I/S$ instead of I would address this, at the cost of an indeterminate form wherever a segment misses the support. (ii) The order is capped near three by the C^0 image interpolant rather than by the multilevel structure. (iii) Zero extension is exact only for compactly supported fields; a Gaussian truncated at the domain boundary at the 10^{-4} level produces a matching error floor.

Extensions. The three-dimensional case is not a difficulty of principle. Concatenation (8) is dimension-independent; the direction grid becomes two-dimensional on the sphere, so (11) becomes an area criterion and the number of directions at level l grows as $(\ell_l/h)^2$, while the number of segment positions falls by the same factor, so the work per level is again constant.

The adjoint $\mathcal{R}_{\mu}^*$, needed by every method in Section 1.2, is the transpose of Algorithm 1 step by step: the merge transposes to a scatter and the angular prolongation to anteroprolongation, in the sense of [6]. The transpose has the same cost, and implementing it is the immediate next step, as is the nonlinear inversion — the FAS/FMG treatment of (5), and of the joint estimation of (f, μ) in which μ enters nonlinearly — sketched in Section 1.2.

Table 4: Accuracy parameters at $n = 129$, $n_\theta = 128$; one varied at a time from the defaults $n_\theta^{\min} = 16$, $\sigma = 2$, $\nu = 2$ and fourth-order stencils. The third column is the constant c of (11), i.e. $\max_l \delta\theta_l \ell_l / h$ over the refining levels. The upper block is the sampling parameters, the lower block the interpolation stencil widths.

knob	value	c	e_{alg}	Gflop
$n_\theta^{\min}$	4	1.571	2.22×10^{-3}	0.69
$n_\theta^{\min}$	8	0.785	5.76×10^{-4}	1.16
$n_\theta^{\min}$	16	0.393	1.38×10^{-4}	1.90
$n_\theta^{\min}$	32	0.196	9.30×10^{-5}	2.98
$n_\theta^{\min}$	64	0.098	9.41×10^{-5}	4.33
σ	1		1.60×10^{-3}	0.95
σ	2		1.38×10^{-4}	1.90
σ	4		7.25×10^{-5}	3.79
ν	2		1.38×10^{-4}	1.90
ν	4		1.38×10^{-4}	3.79
ν	8		1.36×10^{-4}	7.56
q (θ stencil)	2		2.68×10^{-4}	1.29
q (θ stencil)	4		1.38×10^{-4}	1.90
q (θ stencil)	6		1.35×10^{-4}	2.51
p_s (s stencil)	2		1.96×10^{-3}	1.29
p_s (s stencil)	4		1.38×10^{-4}	1.90
p_s (s stencil)	6		1.07×10^{-4}	2.51
p_t (t stencil)	2		8.94×10^{-4}	1.29
p_t (t stencil)	4		1.38×10^{-4}	1.90
p_t (t stencil)	6		1.40×10^{-4}	2.51

7 Conclusion

The attenuated Radon transform admits an $\mathcal{O}(N \log N)$ multilevel evaluation. The construction rests on an exact concatenation identity for the triple (S, E, I) , on interpolating the exponent rather than its exponential and the exit-referenced rather than the midpoint-referenced attenuated integral, and on a level schedule that refines the angular grid at the finest levels where segments are short. On a battery of attenuation fields the error the schedule introduces stays an order of magnitude below the discretisation error of the grid and converges with it, at a cost already several times below direct evaluation at moderate resolution and improving as $\mathcal{O}(\sqrt{N}/\log N)$.

Acknowledgements

The approach follows the multilevel integral-transform framework developed by Achi Brandt and collaborators, and an unpublished scheme for this problem communicated by Achi Brandt to the first author.

Reproducibility

Code, tests and the scripts producing every table and figure are at <https://github.com/orenlivne/tomogrid>. The scaling study is `python -m experiments.scaling`.

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